11–12 第2篇

阅读理解（说明文） Why Is Snow White?

① Every December, when the first heavy snowfall covers our schoolyard, we say without thinking: "**The snow is so white!**" We say it the way we say "the sky is blue" or "fire is hot" — as a simple fact. Yet the whiteness of snow is one of the prettiest tricks in nature. It is neither a paint nor a dye, nor a special kind of cold. It is, in fact, the result of **a little bit of geometry** and **a lot of tiny mirrors**.

② The first truth might surprise you: **a single snowflake is not white; it is clear**. Snowflakes are tiny pieces of **ice**, and ice, like glass, allows light to pass through. If you hold one snowflake on your dark glove and look at it with a magnifying glass, you will see a small **transparent** six-armed star. So where does the whiteness come from? The answer begins when **many snowflakes** gather together — on the ground, on your coat, on the roof of the shed.

③ To understand the white, we must first understand **how we see colour**. Sunlight is not simply "white" light — it is a bundle of many colours: red, orange, yellow, green, blue, indigo, violet. We see each colour when an object **reflects only that colour** back into our eyes. A leaf looks green because it reflects green and absorbs the rest. A red apple looks red because it reflects red. And something looks **white** when it reflects **all the colours of sunlight equally**. That is the secret to the whiteness of snow.

④ Here is where geometry enters. A fresh snowflake has many **flat, tiny surfaces** — edges, points, hollows. When millions of snowflakes pile up together, light cannot go straight through; it hits one surface, bends, bounces off another, hits a third. It is scattered in every direction at every wavelength. Because every colour bounces the same way, nothing is "lost" and nothing is "saved". When our eyes catch these scattered rays, we see **a perfect mixture of all colours** — and our brain calls that mixture **white**.

⑤ The same rule explains many familiar whites in nature. The white of clouds? Millions of tiny water droplets, each one clear, together scatter all colours equally. The white of sea foam? Many tiny bubbles of transparent water, doing the same trick. The white in a glass of milk? Countless little droplets of butter fat inside clear water, each droplet just big enough to bend all colours. The rule is a good friend to remember: **clear pieces + many surfaces = white**.

⑥ There are small but interesting exceptions. **Very deep snow** sometimes appears slightly **blue** when you look into a deep footprint or a snow cave. This is because red light, travelling through a thick pile of ice, is absorbed a little more than blue light. After a long journey inside the deep snow, red has been "eaten" a bit, and only blue returns to

your eye. This is the same reason that the bottom of a very deep pool of clear water can look blue.

⑦ Another gentle exception is **old snow**. Snow that has lain on a busy road for a week is rarely pure white; it is dull grey or brown. This is because the fresh mirrors on each snowflake get covered by city dust, soot, and tiny bits of sand. Dust absorbs some colours instead of reflecting them equally; the snow now looks darker. Fresh snow is bright white because every tiny mirror is still **clean**. Old snow, like an old window, remembers the world it has been next to.

⑧ So the next time a heavy snowfall comes, stand still in your schoolyard for a minute. The whiteness you see is not simply a colour; it is a **story of geometry**. A cloud of transparent stars fell from the sky. They joined together on the ground. Sunlight struck them and was **broken apart, then gathered again**. What your eye sees is the equal return of all the colours of the sky — a quiet, bright mathematics of scattered light. **The world, when clean, is the whitest and widest notebook we have.** Write on it gently; the wind does not like loud walks.

注释： snowflake 雪花；transparent 透明的；reflect 反射；absorb 吸收；scatter 散射

根据短文内容，选择最佳答案。

1. Which statement about a SINGLE snowflake is TRUE?

A. A single snowflake is painted white inside.
B. A single snowflake is clear; snowflakes are tiny pieces of ice and allow light to pass through — if you look at one on a dark glove, you can see a small transparent six-armed star.
C. A single snowflake has no shape.
D. A single snowflake reflects only blue light.

2. According to Paragraph ③, why does an object look WHITE?

A. Because it absorbs all the colours of sunlight.
B. Because it is painted white.
C. Because it reflects all the colours of sunlight equally.
D. Because it stays in the fridge.

3. How do many snowflakes TOGETHER produce the white we see?

A. They share a chemical called "white".
B. Millions of snowflakes have many flat, tiny surfaces; light hits one surface, bends, bounces off another, gets scattered in every direction at every wavelength; every colour bounces the same way, so our eyes see a perfect mixture of all colours, which the brain calls white.

C. They cancel all colours to zero.
D. They only reflect sunlight straight up.

4. Which of the following is NOT an example used in the passage of the same "clear pieces + many surfaces = white" rule?

A. Clouds in the sky.
B. Sea foam.
C. The white in a glass of milk.
D. The yellow of a banana.

5. Why is OLD snow on a busy road often grey or brown, rather than bright white?

A. Because old snow has melted.
B. Because the tiny mirrors on each snowflake get covered by city dust, soot, and sand, which absorb some colours instead of reflecting them equally; old snow, like an old window, remembers the world it has been next to.
C. Because old snow has a different shape.
D. Because the sun sets earlier in winter.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "single snowflake is not white; it is clear ... transparent six-armed star"
【解析】关键事实直引。A、C、D 均错。故选 B。

2.
【答案】C
【题型】细节理解题
【定位】第③段 "reflects all the colours of sunlight equally"
【解析】白色定义。A、B、D 均错。故选 C。

3.
【答案】B
【题型】细节理解题
【定位】第④段散射过程
【解析】几何-光学叙述直引。A、C、D 均错。故选 B。

4.
【答案】D
【题型】细节理解题（否定）
【定位】第⑤段三例云、海浪泡沫、牛奶

【解析】A、B、C 均在文中；香蕉的黄色是吸收其他颜色只反射黄，不符合 "all colours equally" 的规律。故选 D。

5.

【答案】B

【题型】细节理解题

【定位】第⑦段 "dust, soot ... absorb some colours ... like an old window"

【解析】老雪变灰成因直引。A、C、D 均错。故选 B。